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Composable Committed Components

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Composable Committed Components (CCC) in one slide

- Vision of Lehman *et al.*: extend Anonymous Credentials module with additional Zero Knowledge Proof (ZKP) features via “plug and play” modules
- We have generalized this vision and implemented CCC in Rust based on it
- CCC enables declarative specification of graphs using reusable ZKP “components” (nodes) and ZKP “equalities” (edges)
- Advantages of generality include:
  - Enabling privacy-preserving presentation of multiple credentials with different formats and/or signature schemes
  - Reusable components and equalities
  - Convenience, flexibility, finer-grained parallelism
- Open source prototype implementation here

Icon attributions on last slide: 1, 2

Lehman et al. vision

Vision: A Modular Framework for Anonymous Credential Systems

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Abstract. Anonymous credentials enable the unlinkable presentation of previously attested information, or even only predicates thereof. They are a versatile tool and currently enjoy attention in various real-world applications, ranging from the European Digital Identity project to Privacy Pass. While each application usually requires their own tailored variant of anonymous credentials, they all share the same common blueprint. So far, this has not been leveraged though, and currently several proposals either targeting monolithic variants of core components such as BBS signatures, or application-specific protocols undergo standardization. This is clearly not optimal, as the same work gets repeated multiple times, while still risking ending up with many slight modifications of the same main idea and protocols. In this work we present our vision to use a modular approach to build anonymous credential systems: they are built from a core component – consisting of a commitment, signature and NIZK scheme – that can be extended with additional commitment-based modules in a plug-and-play manner. We sketch modules for pseudonyms, range proofs and device binding. Importantly, apart from the committed input, all modules are entirely independent of each other. We use this modularity to propose a concrete instantiation that uses BBS signatures for the core component and ECDSA signatures for device binding, addressing the need to bind modern credential schemes to legacy signatures in secure hardware elements.

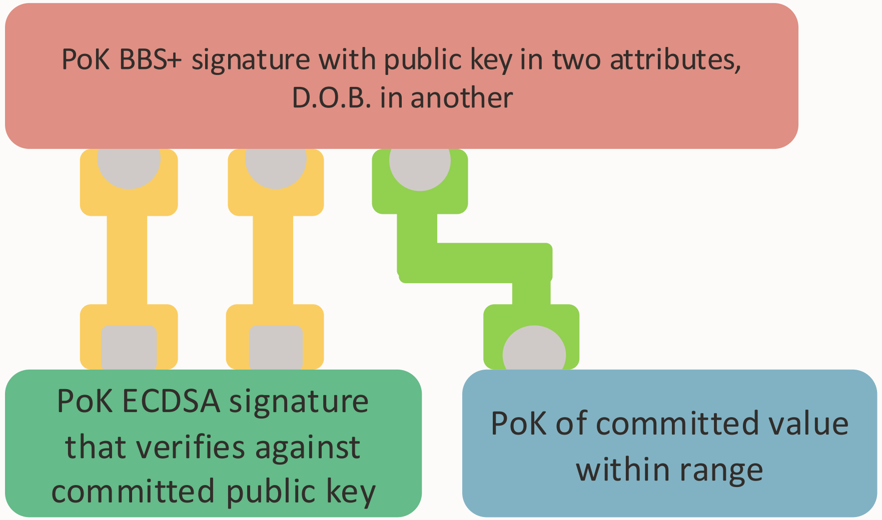
Security Standardisation Research Conference (SSR 2025)   pdf

- Privacy-preserving Anonymous Credentials
- Modular plug-and-play ZKP features
- Enable independent evolution of ZKP mechanisms and commitment schemes
  - e.g., incremental post-quantum safety
- Anonymous Credential module (*CD*) supports selective disclosure and committing to attributes
- Additional modules (R_i) support arbitrary ZKPs over committed values, e.g.,
  - Range proofs (age over 18)
  - PoK of ECDSA signature over committed public key => privacy-preserving device binding of modern Anonymous Credentials to legacy secure elements
  - …

CCC: A prototype implementation of a generalisation of the vision

- No “special” Anonymous Credentials component (like *CD* of Lehman *et al.*):
  - A graph can support zero, one or multiple Anonymous Credential modules
  - Enables presentation of multiple, linked credentials, no need for all Issuers to use same credential format and signature scheme
- Equalities prove that values committed by two commitments are equal (does not require equal commitments)
  - More flexible and convenient (each component can commit independently of others)
  - Commitments can be of different types
  - Enables reuse of equality mechanisms between different components
  - Enables finer-grained parallelism

Example: device binding and range proof



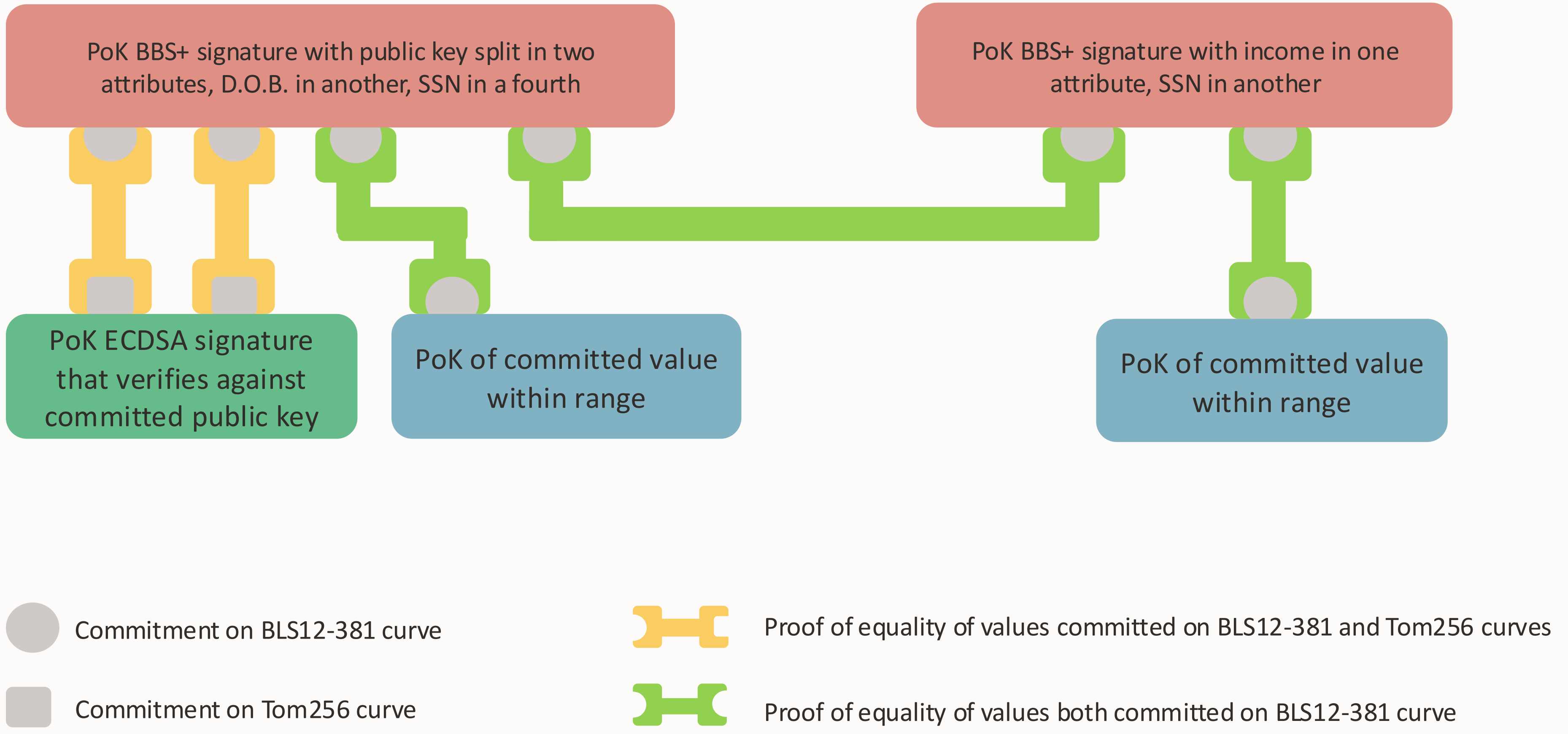
Commitment on BLS12-381 curve

Commitment on Tom256 curve

Proof of equality of values committed on BLS12-381 and Tom256 curves

Proof of equality of values both committed on BLS12-381 curve

Example: device binding, two range proofs, two credentials (same SSN)



Status

- Open source Rust prototype
- In branch `committed_composable_components` of DockNetwork crypto fork; see README
- Examples and tests, including multiple linked, device-bound credentials and a range proof
- Not even reviewed, let alone audited
- Feedback and engagement welcome
- Resources for continuing work unclear

(Aspirations for?) future work

- Our work on VCA (Verifiable Credentials Abstraction) decouples credential formats from underlying ZKP mechanisms
- Enables independent evolution of formats and cryptography. We implemented AnonCreds (v1) over VCA, instantiated with two ZK proof systems: DockNetwork crypto (DNC) and AnonCreds v2 (AC2C); see IIW 41 slides
- VCA + CCC
  - Extend VCA to support committing to attributes
  - Build CCC component supporting VCA
  - Enable (for example), privacy-preserving presentation of a credential in AnonCreds v1 format linked to another in SD-JWT format via a common attribute, e.g., Social Security Number

- “Hybrid” VCA
  - Implement an equality (edge) to prove values committed by AC2C and by DNC equal, enabling “hybrid” VCA implementations via CCC
  - Although both use BLS12-381 curve, may be more challenging than it seems, due to different representations, different ways of encoding raw values to Scalars, etc.

Summary

- Committed Composable Components is an implementation of a generalisation of the vision of Lehman et al.
- Goal: Enable privacy-preserving presentation of multiple linked anonymous credentials, eliminate need for all Issuers to use same credential formats and signature schemes
- Framework enables modular improvement of ZKP mechanisms and commitment schemes, e.g., incremental progress towards post quantum safety
- Initial prototype not reviewed, resources for future work unclear

Thank you

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